

Session - 8

17/9/2026

* Inequalities - 7

• Either or Condition :-

Meaning - Possibility

Possibility (Either-or)

Relation

Non-Relation

- 1) Variable must be same*
- 2) Conclusion Individually false
- 3) Add the Symbols, if the resultant is true ~~after~~ then either or if not then it is neither nor (false).

- 1) There should be no-relation betⁿ variables.
- 2) Conclusion individually false
- 3) Variables same*
- 4) Pick any variable, it should be having ($>$, $<$, $=$) sign only once, if any of the sign is repeated or missing it is neither nor.

Eg-

$$P \geq Q \geq R$$

$$\textcircled{1} P \geq R \rightarrow \text{false}$$

$$\textcircled{2} P = R \rightarrow \text{false}$$

$$\begin{array}{c} \downarrow \\ \hline P \geq R \end{array}$$

$$\rightarrow \text{True}$$

} either or.

1) $A \geq B$; $B \geq C$; $C \geq D$

$$\begin{array}{ll} 1) A > B & F \rightarrow (+) A \geq B \rightarrow T \\ A = B & F \end{array}$$

} either or

$$\begin{array}{ll} 2) B > D & F \rightarrow (+) B \geq D \rightarrow T \\ B = D & F \end{array}$$

} either or

$$\begin{array}{ll} 3) D > B & F \rightarrow (+) D \geq B \rightarrow f \\ D = B & F \end{array}$$

} neither nor

2) $P \geq Q = R$

1) $P > R \rightarrow \text{false}$

2) $P = Q \rightarrow \text{false}$

} either or

$\underline{P \geq Q \mid R} \rightarrow \text{True}$

exceptional case

3) $A \geq B > C$

1) $A > C \rightarrow \text{True}$

only 1 true

2) $A = C \rightarrow \text{false}$

$\underline{A \geq C} \rightarrow \text{false}$

4) $A \geq B ; C \leq D > E = F \geq G$

1) $A > C \quad F \quad \text{neither}$

$A = C \quad F \quad \text{nor}$

$\underline{A \geq C} \quad F$

2) $C > E \rightarrow F \quad \text{neither}$

$C = E \rightarrow F \quad \text{nor}$

$\underline{C \geq E} \rightarrow F$

3) $D > F \quad T \quad \checkmark$

$D = F \quad F \quad X \quad \text{neither}$

~~$D \geq F$~~

4) $E > G \rightarrow F$

$F = G \rightarrow F$

$\underline{E \mid F \geq G} \rightarrow T$

either or

5) $P = Q = R \geq S \geq T = U$

1) $P > Q \quad F$

$P = Q \quad T$

3) $R > T \quad F$

$R = T \quad F$

$\underline{R \geq T} \quad T$

either or

2) $Q > S \quad F$

$R = S \quad F$

$\underline{Q \mid R \geq S} \quad T$

4) $S > T \quad F$

$S = U \quad F$

$\underline{S \geq U \mid T} \quad T$

either or